DSA06 - Data Handling and Visualization | Identify Skewness using Violin Plots (Slot D)

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Rule used: the tail of a violin points to the direction of the skew. Tail to the top (high values) = **right skew**, mean > median. Tail to the bottom (low values) = **left skew**, mean < median. Balanced shape = **symmetric**, mean ~ median. The mean is pulled toward the tail; the median (white dot) stays with the bulk.

Part A (18 marks)

Chart 1: Waiting time (minutes)

Violin	Skew	Mean vs Median	Reason
Ward D	Left skew	Mean < Median	Bulk of data is high; thin tail stretches downward (to low values). (bulk 45-60, tail down to ~23)
Ward B	Right skew	Mean > Median	Bulk of data is low; thin tail stretches upward (to high values). (bulk 5-20, tail up to ~42)
Ward A	Symmetric	Mean ~ Median	Both halves look like mirror images around the white dot.

Chart 2: Rainfall (mm)

Violin	Skew	Mean vs Median	Reason
Station 2	Left skew	Mean < Median	Bulk of data is high; thin tail stretches downward (to low values). (bulk 220-300, tail down to ~60)
Station 3	Right skew	Mean > Median	Bulk of data is low; thin tail stretches upward (to high values). (bulk 0-80, tail up to ~165)
Station 4	Symmetric	Mean ~ Median	Both halves look like mirror images around the white dot.

Chart 3: Order value (Rs hundred)

Violin	Skew	Mean vs Median	Reason
Store 2	Left skew	Mean < Median	Bulk of data is high; thin tail stretches downward (to low values). (bulk 95-120, tail down to ~58)
Store 5	Right skew	Mean > Median	Bulk of data is low; thin tail stretches upward (to high values). (bulk 5-40, tail up to ~85)
Store 3	Symmetric	Mean ~ Median	Both halves look like mirror images around the white dot.

Chart 4: Daily steps (thousand steps)

Violin	Skew	Mean vs Median	Reason
Group 5	Right skew	Mean > Median	Bulk of data is low; thin tail stretches upward (to high values). (bulk 3-7, tail up to ~15.5)
Group 3	Symmetric	Mean ~ Median	Both halves look like mirror images around the white dot.
Group 1	Left skew	Mean < Median	Bulk of data is high; thin tail stretches downward (to low values). (bulk 14-18, tail down to ~6.5)

Part B (2 marks)

B1. In Chart 4, violin 'Group 5' is not symmetric. Towards which side is its tail, and which is the better measure of a typical daily steps here: mean or median? Give one reason.

Answer: The tail of Group 5 is on the **right (upper) side**, towards high step counts (it is right skewed). The **median** is the better measure of typical daily steps, because the few very high values in the long tail pull the mean upward, while the median is not affected by extreme values and stays in the bulk of the data.

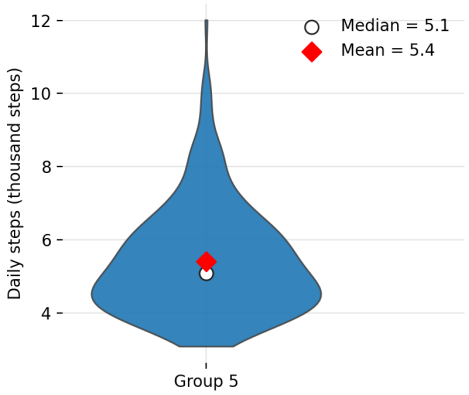


Illustration (simulated data shaped like Group 5): the mean (red diamond) is pulled above the median (white dot) by the right tail.

B2. Write one ggplot2 line that draws violins of value for each group from a data frame df (columns group, value), with the median shown.

Answer (one line):

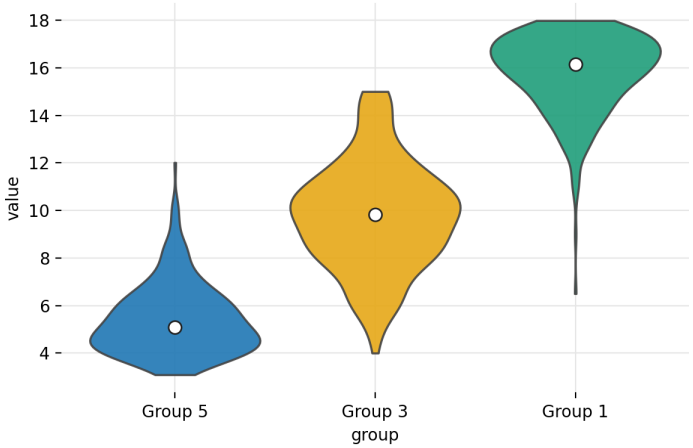
```
ggplot(df, aes(x = group, y = value, fill = group)) + geom_violin() +
  stat_summary(fun = median, geom = "point", shape = 21, fill = "white", size = 3)
```

Explanation: `geom_violin()` draws one violin per group; `stat_summary(fun = median, geom = "point")` adds the median as a white dot on each violin.

Full runnable code (with sample data):

```
library(ggplot2)
set.seed(12)
df <- data.frame(
  group = rep(c("Group 5","Group 3","Group 1"), each = 300),
  value = c(3 + rgamma(300, 2.2, 1/1.1),      # right skewed
            rnorm(300, 9.8, 2.3),             # symmetric
            18.2 - rgamma(300, 2.2, 1))       # left skewed
)
ggplot(df, aes(x = group, y = value, fill = group)) + geom_violin() +
  stat_summary(fun = median, geom = "point", shape = 21, fill = "white", size = 3)
```

Output:



Output of the code: one violin per group with the median shown as a white dot. (Preview drawn with sample data; running the R code in RStudio gives the same style of plot.)